

Technology Stack

Date	11 July 2026
Team ID	
Project Name	Rising waters - Flood Prediction using Machine Learning
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	<i>HTML5, CSS3, JavaScript</i>	<i>Used to build a simple, responsive, and user-friendly interface for entering weather parameters and displaying flood prediction results.</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python, Flask</i>	<i>Flask provides a lightweight web framework that integrates easily with the machine learning model and processes prediction requests efficiently.</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>CSV Dataset(Rainfall & Weather Data)</i>	<i>Stores historical rainfall and weather data used for training and testing the machine learning model.</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Render</i>	<i>Hosts the Flask application online, allowing users to access the flood prediction system from anywhere.</i>
5	Version Control & CI/CD	<i>Git, GitHub</i>	<i>Used for source code management, version control, and team collaboration during project development.</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Scikit-learn, PAndas, NumPy, Matplotlib</i>	<i>Scikit-learn is used for machine learning models, Pandas for data preprocessing, NumPy for numerical computations, and Matplotlib for data visualization.</i>